

Chaos-enhanced accelerated particle swarm optimization

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ARTICLE INFO

Article history:

Received 27 December 2011

Received in revised form 18 May 2012

Accepted 17 July 2012

Available online 31 July 2012

Keywords:

Accelerated particle swarm optimization

Chaotic maps

Global optimization

ABSTRACT

There are more than two dozen variants of particle swarm optimization (PSO) algorithms in the literature. Recently, a new variant, called accelerated PSO (APSO), shows some extra advantages in convergence for global search. In the present study, we will introduce chaos into the APSO in order to further enhance its global search ability. Firstly, detailed studies are carried out on benchmark problems with twelve different chaotic maps to find out the most efficient one. Then the chaotic APSO (CAPSO) will be compared with some other chaotic PSO algorithms presented in the literature. The performance of the CAPSO algorithm is also validated using three engineering problems. The results show that the CAPSO with an appropriate chaotic map can clearly outperform standard APSO, with very good performance in comparison with other algorithms and in application to a complex problem.

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1. Introduction

Nonlinearity of many design optimization problems often results in multiple local optima, which poses substantial challenges in finding the global optimality of interest. To cope with this issue, global algorithms are widely used [1]. Metaheuristic techniques are well-known global optimization methods. These techniques attempt to mimic some characteristics of natural phenomena or social behavior [2].

Several novel metaheuristic algorithms have been proposed for global optimization. Such algorithms can increase the overall computational efficiency, solve large-scale problems, and implement robust optimization codes [2]. Swarm intelligent algorithms, such as particle swarm optimization (PSO) and krill herd algorithms [3], are one of the main categories of the metaheuristic algorithms. Recently, Yang proposed an improved PSO algorithm, called the accelerated PSO (APSO) [4]. Preliminary studies suggest that the APSO can perform superiorly, compared with genetic algorithms (GA) and PSO [4].

Intensification and diversification are two important characteristics of metaheuristic methods [5]. Intensification searches around the current best solutions and selects the best candidate designs, while diversification allows the optimizer to explore the search space more efficiently, mostly by means of randomization. The important role of randomizing can be played by using chaos theory. Previously, chaotic sequences have been used together with some heuristic optimization algorithms to produce hybrid algorithms, and such hybridization has been successful in combination with GA [6], harmony search (HS) [7], simulated annealing (SA) [8], imperialist competitive algorithm [9] and firefly algorithm (FA) [10]. Different chaotic PSO (CPSO) variants have also been proposed by researchers [11–15] and the results have indicated good performance of the chaotic PSO variants.

This paper introduces a new chaos-enhanced accelerated particle swarm optimization, or chaotic APSO (CAPSO). We can use different chaotic maps to enhance different aspects of the algorithm, and thus lead to a class of such algorithms. In these

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algorithms, we will focus on using different chaotic systems to replace the main parameters of the APSO. Thus different methods that use chaotic maps as efficient alternatives to pseudorandom sequences have been proposed. In order to evaluate the proposed algorithms and find the best chaotic map APSO, we will use a set of well-chosen, well-known mathematical benchmarks in the literature. The results reveal the noticeable improvements of the new algorithms due to the application of deterministic chaotic signals in place of constant values. Then, CAPSO is compared with some other CPSO variants proposed in the literature. To evaluate the performance of the proposed CAPSO over engineering problems, vibration-based damage detection problems in three scenarios, gear train design, and pressure vessel design, have been solved.

2. Accelerated particle swarm optimization

2.1. Review of particle swarm optimization

The standard particle swarm optimization (PSO) was developed by Kennedy and Eberhart in 1995 [16], inspired by the swarm behavior such as fish and bird schooling in nature. Extensive studies suggest that PSO is very efficient for many optimization problems; however, sometimes, it may suffer from premature convergence for highly multimodal problems. In the literature, there are at least two dozens PSO variants, which have been formed mainly by slight variations of certain terms and parameters in the PSO, especially by introducing inertia parameter and by hybridizing with other algorithms.

In the standard PSO, particles interact and form zig-zag trajectories during the search process. Each particle moves with two components: a deterministic component and a stochastic component. That is, each particle is attracted toward the position of the current global best g^* and its own best location x_i^* in history, while at the same time it has a tendency to move randomly.

For a particle with position vector $\mathbf{x}_i$ and velocity $\mathbf{v}_i$, its velocity at a new time step is updated as

$$\mathbf{v}_i^{t+1} = \mathbf{v}_i^t + \alpha r_1 (\mathbf{x}_i^* - \mathbf{x}_i^t) + \beta r_2 (\mathbf{g}^* - \mathbf{x}_i^t) \quad (1)$$

where r_1 and r_2 are two random vectors drawn from a uniform distribution. Here, parameters α and β are often called the learning parameters or acceleration constants, which can typically be taken as, say, $\alpha \approx \beta \approx 2$.

At a new position during iterations, the position vector of a particle is updated by

$$\mathbf{x}_i^{t+1} = \mathbf{x}_i^t + \mathbf{v}_i^{t+1} \Delta t \quad (2)$$

where Δt , time interval, is equal unit.

Among more than two dozen variants of PSO, the most noticeable improvement is possibly the use of an inertia function $\theta(t)$ so that $\mathbf{v}_i^t$ is replaced by $\theta(t) \mathbf{v}_i^t$

$$\mathbf{v}_i^{t+1} = \theta(t) \mathbf{v}_i^t + \alpha r_1 (\mathbf{x}_i^* - \mathbf{x}_i^t) + \beta r_2 (\mathbf{g}^* - \mathbf{x}_i^t) \quad (3)$$

where $\theta \in (0, 1)$ [17]. In the simplest case, the inertia function can be taken as a constant, typically $\theta \approx 0.5$ to 0.9 , though the best scheme is to use a higher value at first and then gradually reduce it to a slightly lower value during iterations. From the analogy in physical systems, increase of mass often stabilizes the motion of the system. Similarly, the introduction of inertia is essentially to use a virtual mass to stabilize the movements of particles, thus leading to potential speed up in convergence.

2.2. Convergence analysis

Convergence analysis is an important topic, though there are only limited results concerning metaheuristic algorithms. For example, Clerc and Kennedy [17] provided a detailed analysis of PSO using a simplified dynamical system by considering a 1D particle system with a global best p and transforming $\lambda = (\beta + \alpha)$ and $u_t = p - x(t+1)$. This essentially leads to

$$Y_{t+1} = AY_t \quad (4)$$

where

$$A = \begin{bmatrix} 1 & \lambda \\ -1 & 1 - \lambda \end{bmatrix} \quad Y_t = \begin{Bmatrix} v_t \\ u_t \end{Bmatrix}$$

In general, the solution to this dynamical system can be characterized by the eigenvalues of

$$\lambda_{1,2} = 1 - \frac{\lambda}{2} \pm \frac{\sqrt{\lambda^2 - 4\lambda}}{2} \quad (5)$$

Critically, $\lambda = 4$ leads to a bifurcation. Some straightforward analysis shows three cases. For $0 < \lambda < 4$, cyclic and/or quasi-cyclic trajectories exist, while for $\lambda > 4$, non-cyclic behavior can be expected and the distance from Y_t to the center $(0,0)$ is monotonically increasing with t . In a special case $\lambda = 4$, convergence behavior can be observed [16].

2.3. Accelerated particle swarm optimization

The standard PSO uses both the current global best g^* and the individual best x_i^* . One of the reasons of using the individual best is primarily to increase the diversity in the quality solutions; however, this diversity can be simulated or achieved by using some randomness. Subsequently, there is no compelling reason for using the individual best, unless the optimization problem of interest is highly nonlinear and multimodal [3].

A simplified version that could accelerate the convergence of the algorithm is to use the global best only. In 2008, Yang developed an accelerated particle swarm optimization (APSO) [1], and the velocity vector is generated by a simpler formula as:

$$v_i^{t+1} = v_i^t + \alpha r(t) + \beta(g^* - x_i^t) \quad (6)$$

where r is drawn from $N(0, 1)$ to replace the second term. Here this update of the position is essentially the same as Eq. (2); however, we can see later that the velocity is redundant in accelerated PSO. In order to increase the convergence even further, we can also write the update of the location in a single step, as

$$x_i^{t+1} = (1 - \beta)x_i^t + \beta g^* + \alpha r \quad (7)$$

This simpler version will give the same order of convergence. It is worth pointing out that the velocity does not appear in Eq. (7), and thus there is no need to deal with the initialization of velocity vectors. Therefore, the APSO is much simpler to implement. Here the randomization term αr provides the ability for the system to escape any local optima if α is chosen to be consistent with the scales of the problem of interest, while r can be drawn from a probability distribution such as Gaussian. Typically, $\alpha = 0.1L$ to $0.5L$ where L is the scale of each variable, while $\beta = 0.2$ to 0.7 is sufficient for most applications.

Comparing with many PSO variants, the APSO uses only two parameters (α and β), and the mechanism is simple to understand. A further improvement to the APSO is to reduce the randomness as iterations proceed. This means that we can use a monotonically decreasing function. In our implementation, we use

$$\alpha = \delta^t \quad (8)$$

where $0 < \delta < 1$ is an annealing-like parameter whose value can be taken as 0.7 to 0.9 for most cases. Here $t \in [0, t_{max}]$ and t_{max} is the maximum of iterations.

2.4. Convergence analysis of APSO

Using a dynamical approach similar to that in Section 2.2, ignoring the randomness term and setting $u = g^* - x_i^t$, we have a system

$$v_i^{t+1} = v_i^t + \beta u_i^t, \quad u_i^{t+1} = (1 - \beta)u_i^t \quad (9)$$

or

$$Y_{t+1} = AY_t, \quad A = \begin{bmatrix} 1 & \beta \\ 0 & 1 - \beta \end{bmatrix},$$

Therefore, its eigenvalues are simply 1 and $1 - \beta$. From theory of dynamical systems, the above iterations will always converge stably for any $\beta > 0$. In fact, our simulations also confirm that APSO converges very quickly and stably.

2.5. Chaos-enhanced accelerated particle swarm optimization

As it is presented in Eq. (7), the main parameter of the APSO is the attraction parameter β that characterizes the variations of the global best attraction, and its value is crucially important in determining the speed of the convergence and how the APSO behaves.

Based on a parametric study by varying β from 0 to 1 by a step of 0.1, and we found that β should be in $[0.2, 0.7]$ for most problems.

The basic APSO is very efficient, but the solutions are still changing slightly as the optima are approaching. According to the previously used and suggested values in [18], we used $\beta = 0.5$ for the standard APSO.

From Eq. (7), we can see that the particles will converge to the global best when $\beta = 1$ in any given step, which will remain stationary even if the current global best may not be the true global best. On the other hand, the algorithm may lead to slow changes when $\beta = 0$. Therefore, a well-tuned β is very important.

In the standard APSO, there is no need to keep β constant. In fact, a varying β may be advantageous, which may also lead the speedup of convergence of the algorithm as we will see in the rest of this paper. As all chaotic maps are normalized, the variations of a chaotic map are always between $[0, 1]$. Therefore, chaotic maps can be used to tune parameter β , and such chaos-enhanced APSO is referred as the chaotic APSO.

The pseudo code of a schematic chaotic APSO is presented in Fig. 1.

Chaotic APSO

```

Initialize locations  $x_i$  of  $n$  particles and chaotic value,  $c_0$ 
Find  $g^*$  at  $t = 0$ 
while (criterion )
    Update the iteration counter,  $t = t + 1$ 
    Update the attraction parameter,  $\beta = c_{t+1}$ 
    for  $i = 1:n$ 
        Update locations using equation (7)
        Evaluate objective functions at new locations  $x^{t+1}$ 
    end
    Find the current global best  $g^*$ 
end while
Output the final results

```

Fig. 1. Simple pseudo code of a chaotic APSO.

The tuning of β using chaotic maps also enhance the ability of the APSO to escape local optima in a multimodal landscape, and thus increase the possibility of finding the global optimality in a limited number of iterations. Comparing with other algorithms, this could be an advantage for this algorithm as we can see in the simulations below.

3. Chaotic maps

In the rest of this paper, we will introduce a chaotic-based APSO method. Therefore, we will first review a diverse variety of one-dimensional, non-invertible maps, then we will use them to enhance APSO for optimization.

3.1. Chebyshev map

The family of Chebyshev map is defined as follows [19]:

$$x_{k+1} = \cos(k \cos^{-1}(x_k)) \quad (10)$$

3.2. Circle map

The circle map [20] is represented by the following equation:

$$x_{k+1} = x_k + b - (a/2\pi) \sin(2\pi x_k) \bmod(1) \quad (11)$$

with $a = 0.5$ and $b = 0.2$, it generates chaotic sequence in $(0, 1)$ where $\bmod(1)$ is the remainder of division of the number by 1.

3.3. Gauss/mouse map

The following equations define Gaussian map [21]:

$$x_{k+1} = \begin{cases} 0 & x_k = 0 \\ 1/x_k \bmod(1) & \text{otherwise} \end{cases} \quad (12)$$

$$1/x_k \bmod(1) = \frac{1}{x_k} - \left\lceil \frac{1}{x_k} \right\rceil \quad (13)$$

This map also generates chaotic sequences in $(0, 1)$.

3.4. Intermittency map

The Intermittency map has two parts: one is linear and the other one is nonlinear. It is formulated as [22]:

$$x_{k+1} = \begin{cases} \varepsilon + x_k + c x_k^n & 0 < x_k \leq P \\ \frac{x_k - P}{1 - P} & P < x_k < 1 \end{cases} \quad (14)$$

3.5. Iterative map

The iterative chaotic map with infinite collapses [23] is expressed by:

$$x_{k+1} = \sin\left(\frac{a\pi}{x_k}\right) \quad (15)$$

where $a \in (0, 1)$ is a suitable parameter.

3.6. Liebovitch map

This map is introduced by Liebovitch and Toth [19]. This map can be formulated as

$$x_{k+1} = \begin{cases} \alpha x_k & 0 < x_k \leq P_1 \\ \frac{P-x_k}{P_2-P_1} & P_1 < x_k \leq P_2 \\ 1 - \beta(1 - x_k) & P_2 < x_k \leq 1 \end{cases} \quad (16)$$

where $\alpha < \beta$ and they are defined as follows:

$$\alpha = \frac{P_2}{P_1}(1 - (P_2 - P_1)) \quad (17)$$

$$\beta = \frac{1}{P_2 - 1}((P_2 - 1) - P_1(P_2 - P_1)) \quad (18)$$

Here three equal limits are considered for the map.

3.7. Logistic map

The logistic map is represented by the following Eq. (19). The equation appears in nonlinear dynamics of biological population evidencing chaotic behavior [22].

$$x_{k+1} = ax_k(1 - x_k) \quad (19)$$

In this equation, x_k is the k th chaotic number, with k denoting the iteration number. Obviously, $x \in (0, 1)$ under the conditions that the initial $x_0 \in (0, 1)$ and that $x_0 \notin \{0.0, 0.25, 0.75, 0.5, 1.0\}$. $a = 4$ is used for the experiments.

3.8. Piecewise map

The piecewise map is defined as follows [23]:

$$x_{k+1} = \begin{cases} \frac{x_k}{P} & 0 \leq x_k < P \\ \frac{x_k - P}{0.5 - P} & P \leq x_k < \frac{1}{2} \\ \frac{1 - P - x_k}{0.5 - P} & \frac{1}{2} \leq x_k < 1 - P \\ \frac{1 - x_k}{P} & 1 - P \leq x_k < 1 \end{cases} \quad (20)$$

where P is the control parameter between 0 and 0.5 and $x \in (0, 1)$.

3.9. Sine map

The sine map is a unimodal map and can be formulated as [24]:

$$x_{k+1} = \frac{a}{4} \sin(\pi x_k) \quad (21)$$

where $0 < a \leq 4$.

3.10. Singer map

Singer map is a one-dimensional system as given below [25]:

$$x_{k+1} = \mu(7.86x_k - 23.31x_k^2 + 28.75x_k^3 - 13.302875x_k^4) \quad (22)$$

where μ is a parameter between 0.9 and 1.08.

3.11. Sinusoidal map

This iterator [26] is as given as follows:

$$x_{k+1} = ax_k^2 \sin(\pi x_k) \quad (23)$$

For $a = 2.3$ and $x_0 = 0.7$ it has the following simplified form:

$$x_{k+1} = \sin(\pi x_k) \quad (24)$$

3.12. Tent map

The tent map is very similar to the logistic map. It displays some very specific chaotic effects. This map is defined by the following equation [27]:

$$x_{k+1} = \begin{cases} \frac{x_k}{0.7} & x_k < 0.7 \\ \frac{10}{3}(1 - x_k) & x_k \geq 0.7 \end{cases} \quad (25)$$

4. Implementation

4.1. Benchmark numerical experiments

Different chaotic APSO variants were benchmarked using three well-known numerical examples, namely Griewank, Ackley and Sphere functions. The formulation of these benchmark functions are presented in Table 1.

4.2. Criterion of success

The success rate can be used as a useful criterion for evaluating the performance of the algorithms which is defined as:

$$S_r = 100 \times \frac{N_{\text{successful}}}{N_{\text{all}}} \quad (26)$$

where N_{all} is the number of all trials, and $N_{\text{successful}}$ is the number of trials in which the solution is successfully found. A successful run can be defined as:

$$\|g^* - X^*\| \leq (UB - LB) \times 10^{-4} \quad (27)$$

where g^* is the obtained global best by the proposed algorithms; UB and LB are, respectively, upper and lower bounds.

4.3. Initialization and parameter studies

We also used 100 different runs for each setting with completely different initial conditions so as to produce meaningful statistics. The final results are found to be almost independent of the initial guess. Also, statistical measures such as mean objective values and their standard deviations are used to measure the performance of the algorithms. This approach is reflected in many tables provided in the main text.

For most cases in our implementation, we have carried out some extensive sensitivity studies of parameters such as the population size. From our simulations, we observed that the population size $n = 40$ will be sufficient for most problems. With a fixed number of n particles at each run, the benchmark functions are optimized within 250 iterations or generations.

5. Simulation results

5.1. Tuning learning parameter, β , with chaotic maps

It is possible to improve the solution quality by using the chaotic maps. In this section, tuning the global best attraction parameter is performed. Here, the value of β is replaced with different chaotic maps introduced in Section 3. In order to fulfill this, all the maps are normalized between 0.2 and 0.7. The statistical results such as best, mean, median, worst and standard

Table 1
Mathematical benchmark problems.

Name	Dimension	Formulation	Bounds
Griewank	20	$f(X) = 1 + \frac{1}{4000} \sum_{i=1}^n x_i^2 - \prod_{i=1}^n \cos(\frac{x_i}{\sqrt{i}})$	$X \in [-600, 600]^{20}$
Ackley	30	$f(X) = -ae^{-0.02\sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}} - e^{n-1} \sum_{i=1}^n \cos(2\pi x_i) + a + e$, $a = 20$	$X \in [-32, 32]^{30}$
Sphere	50	$f(X) = \ X\ $ where $\ X\ = \sqrt{\sum_{i=1}^n x_i^2}$	$X \in [-100, 100]^{40}$

The global optimums are located at the origin ($X^* = 0, 0, \dots, 0$) with $f(X^*) = 0$.

deviation of runs obtained by each algorithm for the Griewank, Ackley and Sphere functions are respectively presented in Tables 2, 3 and 4.

To compare the chaotic APSO with standard APSO and to search for the most effective chaotic map, the success rate of the chaotic APSO is presented in Table 5. As it can be seen, replacing β with chaotic maps can definitely improve the performance of the APSO.

From Table 5, it can be seen that the sinusoidal and singer maps are more effective than others. For these two maps, the Sinusoidal slightly outperforms the Singer map. Therefore, it is selected as the best map to be used for chaotic APSO (CAPSO) which will be compared and discussed in more detail in the next sections.

It is worth pointing out that different chaotic maps may perform differently, but for a given application, there is no unique best chaotic map to recommend as there are often a few chaotic maps that can perform almost equally well. Therefore, it is not easy to choose a chaotic map without actual experimenting. However, from our observations, we find that chaotic maps,

Table 2Statistical results for Griewank function with different β values.

β	Best	Mean	Median	Worst	Std. Dev.
Standard value	5.88×10^{-7}	2.39×10^{-2}	1.97×10^{-2}	9.31×10^{-2}	1.95×10^{-2}
Gauss/mouse map	6.33×10^{-7}	1.95×10^{-2}	1.23×10^{-2}	1.18×10^{-1}	1.92×10^{-2}
Circle map	5.99×10^{-7}	1.87×10^{-2}	1.38×10^{-2}	9.85×10^{-2}	1.72×10^{-2}
ICMIC map	7.60×10^{-7}	2.43×10^{-2}	1.97×10^{-2}	1.59×10^{-1}	2.29×10^{-2}
Logistic map	8.28×10^{-7}	2.13×10^{-2}	1.97×10^{-2}	8.10×10^{-2}	1.85×10^{-2}
Piecewise map	6.31×10^{-7}	2.19×10^{-2}	1.72×10^{-2}	1.23×10^{-1}	2.08×10^{-2}
Sinusoidal map	4.49×10^{-7}	2.00×10^{-2}	1.23×10^{-2}	9.31×10^{-2}	2.12×10^{-2}
Singer map	5.50×10^{-7}	2.59×10^0	1.72×10^{-2}	2.49×10^2	2.49×10^1
Tent map	8.17×10^{-7}	2.21×10^{-2}	1.36×10^{-2}	1.35×10^{-1}	2.36×10^{-2}
Liebovitch map	8.91×10^{-7}	2.42×10^{-2}	1.63×10^{-2}	1.40×10^{-1}	2.69×10^{-2}
Intermittenc map	1.16×10^{-6}	2.43×10^{-2}	1.72×10^{-2}	2.75×10^{-1}	3.48×10^{-2}
Sine map	9.06×10^{-7}	2.18×10^{-2}	1.72×10^{-2}	1.25×10^{-1}	2.15×10^{-2}
Chebyshev map	4.67×10^{-3}	1.39×10^{-1}	9.17×10^{-2}	1.07×10^0	1.81×10^{-1}

Table 3Statistical results for Ackley function with different β values.

β	Best	Mean	Median	Worst	Std. Dev.
Standard value	1.76×10^{-4}	1.71×10^0	1.78×10^0	3.73×10^0	6.96×10^{-1}
Circle map	2.00×10^{-4}	1.89×10^0	1.78×10^0	4.12×10^0	7.26×10^{-1}
Gauss/mouse map	1.97×10^{-4}	1.88×10^0	1.90×10^0	4.74×10^0	8.54×10^{-1}
ICMIC map	1.51×10^{-4}	1.76×10^0	1.71×10^0	3.28×10^0	7.18×10^{-1}
Logistic map	1.72×10^{-1}	1.87×10^0	1.90×10^0	3.28×10^0	6.58×10^{-1}
Piecewise map	2.52×10^{-4}	1.82×10^0	1.90×10^0	4.03×10^0	7.41×10^{-1}
Sinusoidal map	1.71×10^{-4}	1.65×10^0	1.65×10^0	5.16×10^0	8.20×10^{-1}
Singer map	1.85×10^{-4}	1.51×10^0	1.42×10^0	6.08×10^0	1.04×10^0
Tent map	1.95×10^{-4}	1.81×10^0	1.78×10^0	5.47×10^0	8.63×10^{-1}
Liebovitch map	2.33×10^{-4}	1.73×10^0	1.78×10^0	3.46×10^0	7.29×10^{-1}
Intermittenc map	1.24×10^{-1}	2.05×10^0	1.98×10^0	4.62×10^0	7.48×10^{-1}
Sine map	2.27×10^{-4}	1.98×10^0	2.01×10^0	4.17×10^0	7.26×10^{-1}
Chebyshev map	2.38×10^0	4.76×10^0	4.22×10^0	1.30×10^1	1.82×10^0

Table 4Statistical results for Sphere function with different β values.

β	Best	Mean	Median	Worst	Std. Dev.
Standard value	1.70×10^{-6}	1.49×10^{-1}	1.34×10^{-4}	6.01×10^0	7.53×10^{-1}
Circle map	1.14×10^{-6}	5.34×10^1	1.91×10^0	5.40×10^2	1.07×10^2
Gauss/mouse map	9.10×10^{-3}	2.48×10^1	9.72×10^0	1.85×10^2	3.53×10^1
ICMIC map	1.91×10^{-6}	2.67×10^{-1}	2.07×10^{-3}	5.79×10^0	8.61×10^{-1}
Logistic map	3.18×10^{-3}	2.65×10^1	6.82×10^0	4.48×10^2	5.70×10^1
Piecewise map	3.80×10^{-3}	1.31×10^1	2.23×10^0	1.81×10^2	3.00×10^1
Sinusoidal map	8.48×10^{-7}	1.12×10^{-2}	2.59×10^{-6}	1.05×10^0	1.05×10^{-1}
Singer map	1.19×10^{-6}	5.34×10^2	2.52×10^{-6}	5.33×10^4	5.33×10^3
Tent map	1.62×10^{-4}	1.13×10^1	2.69×10^0	2.88×10^2	3.17×10^1
Liebovitch map	3.82×10^{-5}	9.19×10^0	1.15×10^0	1.21×10^2	1.92×10^1
Intermittenc map	1.08×10^{-3}	5.83×10^1	1.82×10^1	7.02×10^2	1.05×10^2
Sine map	1.59×10^{-4}	2.60×10^1	7.74×10^0	2.59×10^2	4.84×10^1
Chebyshev map	1.90×10^2	1.54×10^3	1.14×10^3	5.40×10^3	1.17×10^3

Table 5

Success rates of APSO using the standard value and different chaotic maps.

β	Griewank (20 D)	Ackley (30 D)	Sphere (40 D)
Standard value	10	4	59
Circle map	18	2	38
Gauss/mouse map	16	5	0
ICMIC map	12	6	36
Logistic map	14	0	0
Piecewise map	18	4	0
Sinusoidal map	23	9	92
Singer map	19	10	91
Tent map	17	6	2
Liebovitch map	21	3	4
Intermittenc map	14	0	0
sine map	17	1	1
Chebyshev map	0	0	0

Table 6

Comparison of the CAPSO with five different CPSO variants (CPSO-I to V) proposed in the literature.

References	Method	Griewank	Ackley	Sphere	Rosenbrock	Rastrigrin	Schaffer
[11]	CPSO-I	7.50×10^{-3}	2.80×10^{-1}	4.05×10^{-79}	1.25×10^1	4.73×10^1	0.00×10^0
[12]	CPSO-II	4.33×10^0	1.46×10^1	3.68×10^2	4.92×10^4	1.58×10^2	5.80×10^{-3}
[13]	CPSO-III	3.10×10^{-2}	8.57×10^0	1.90×10^{-3}	3.28×10^2	4.30×10^1	7.50×10^{-6}
[14]	CPSO-IV	4.20×10^{-2}	3.20×10^{-1}	5.25×10^{-77}	1.30×10^1	1.18×10^1	2.90×10^{-3}
[15]	CPSO-V	5.90×10^{-2}	1.30×10^0	7.80×10^{-4}	1.31×10^2	5.64×10^1	1.60×10^{-3}
Present	CAPSO	5.98×10^{-4}	1.20×10^0	1.17×10^{-218}	1.93×10^1	5.66×10^1	0.00×10^0

with a unimode centered around the middle of maps, tend to produce better results, and Singer's map and sinusoidal map fall into this category, and they are indeed very effective.

5.2. Comparison study

There are several chaotic PSO algorithms proposed in the literature. Five of these methods are used here for comparison. These, CPSO-I to V, are respectively suggested by [11–15]. Xiang et al. [11], and Meng et al. [12] add a chaotic search to the PSO algorithm. It should be noticed, Chen et al. [10] used piecewise map and Xiang et al. [11] and Meng et al. [12] utilized Logistic map in their algorithms. Liu et al. [13] and Jiang and Bompard [14] tried to improve PSO algorithm by tuning its parameters using chaotic maps. They have tried to tune the velocity updating [13] and acceleration constants [14]. The performance of the proposed CAPSO is compared with these five algorithms using six well-known benchmarks: Griewank, Ackley, Sphere, Rosenbrock, Rastrigrin and Schaffer functions. The global optimums of these benchmarks are equal to zero. The Detailed descriptions of these functions can be found in [11]. It should be noticed that the population size and number of iteration are, respectively, 30 and 5000 for these simulations. Therefore, they can be compared according to the oracle point of view by only using the best results [28]. According to the simulation results, presented in Table 6, it can be seen that the CPSO-I and CAPSO have the best performance among these algorithms.

6. Engineering benchmark

In this section, three engineering design benchmarks are solved to evaluate the capability of the CAPSO. The engineering problems are structural model updating, gear train design, and pressure vessel design problems. The model updating problem is a complex and continues engineering problem and here, it is solved in three different cases. Then, the gear train design problem is solved to show the capability of the CAPSO to dealing with discrete problem. Finally, the well-known pressure vessel design problem is solved a benchmark mixed-variable problem.

6.1. Model updating problem

This problem is the damage detection of the structures using dynamic responses. Here, the modal flexibility matrix is employed as the objective function, based on previous successes with this approach. For example, Yun et al. [29,30] used a modal flexibility matrix directly obtained from modal test data for damage localization and quantification. The modal flexibility matrix can be formulated as a linear equation of motion:

$$K\Phi = M\Phi\Lambda \quad (28)$$

where Φ is the mode shape matrix and Λ represents the diagonal matrix containing the eigenvalues. Due to orthogonality properties of the mode shape with respect to stiffness and mass matrices, the following relationships are established

$$\Phi^T K \Phi = \Lambda \quad \text{and} \quad \Phi^T M \Phi = I \quad (29)$$

where I is the unity matrix. From Eq. (29), the stiffness matrix can be rewritten as

$$K = \Phi^{-T} \Lambda \Phi^{-1} = (\Phi \Lambda^{-1} \Phi^{-T})^{-1} \quad (30)$$

Proceeding with the relationship between the stiffness and the flexibility matrix, $K = F^{-1}$, the modal flexibility matrix can be derived as

$$F = \Phi \Lambda^{-1} \Phi^{-T} \quad (31)$$

where the modal and eigenvalue matrices can be calculated from the analytical model with a vector of damage parameters. The difference between the tested model and the analytical model can then be utilized as an objective function for damage quantification, as

$$\Pi(\theta) = \|F_{Exp.} - F_{Updated}(\theta)\|_{Fro} \quad (32)$$

where θ is the vector depicting the damage parameters, the sign $\| \cdot \|_{Fro}$ represents the Frobenius norm for the residual matrix. Here $F_{Exp.}$ indicates the modal flexibility of the real structure, and $F_{Updated}$ is the modal flexibility matrix for a particular damage scenario.

One advantage of using the modal flexibility matrix as an objective function is that the modal flexibility matrix is insensitive to high frequency modes because there is an inverse relationship between the flexibility matrix and eigenvalues of the dynamic system. Moreover, unlike the flexibility matrix, the calculation of the modal flexibility matrix is numerically advantageous over that of flexibility matrix since it does not require the inverse of the modal matrix. With no inverse matrix involved, the modal flexibility matrix calculation process is not hindered by large computation times.

Here, an I-section steel beam is selected for evaluation as shown in Fig. 2. This beam can be modeled as an Euler–Bernolli beam. The material and sectional properties of the beam are presented in Table 7.

The I-beam is divided to 10 elements for finite element analysis as shown in Fig. 3. A damaged element is modeled using reduction of its elastic modulus, E . Therefore, the percentage of damage is the percentage of the reduction. Here, three different damage scenarios are defined for evaluation of the CAPSO. In the first and second scenarios, single damage site happened in element No. 4, but the percentages of the damage are different and they are 5% and 20%, respectively. The third damage scenario is a multi-damage one. In this scenario, the damage in element No. 3 and 6 is 5% and 10%, respectively.

For these three cases, the number of search iterations is equal to 3000. The results of damage detection and the related parameters as well as convergence history over the 50 runs are summarized in Figs. 4, 5 and 6 for the three cases, respec-

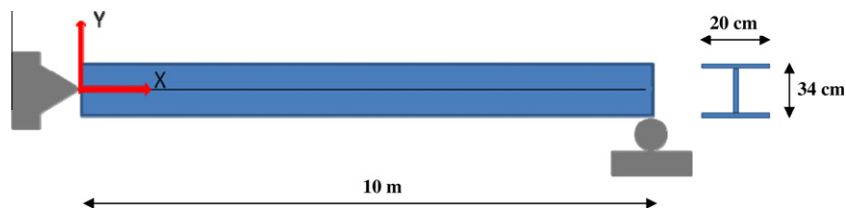


Fig. 2. Schematic representation of I-section beam.

Table 7
Material and sectional properties of the I-beam.

Material properties	Sectional Properties
$E = 210 \times 10^9$ (Pa)	$A = 0.0011$ (m ²)
$\mu = 0.29$	$I = 2.0491 \times 10^{-5}$ (m ⁴)
$\rho = 7850$ (kg/m ³)	$J = 2.3158 \times 10^{-5}$ (m ⁴)

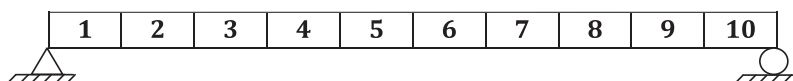


Fig. 3. Element numbering in the I-beam.

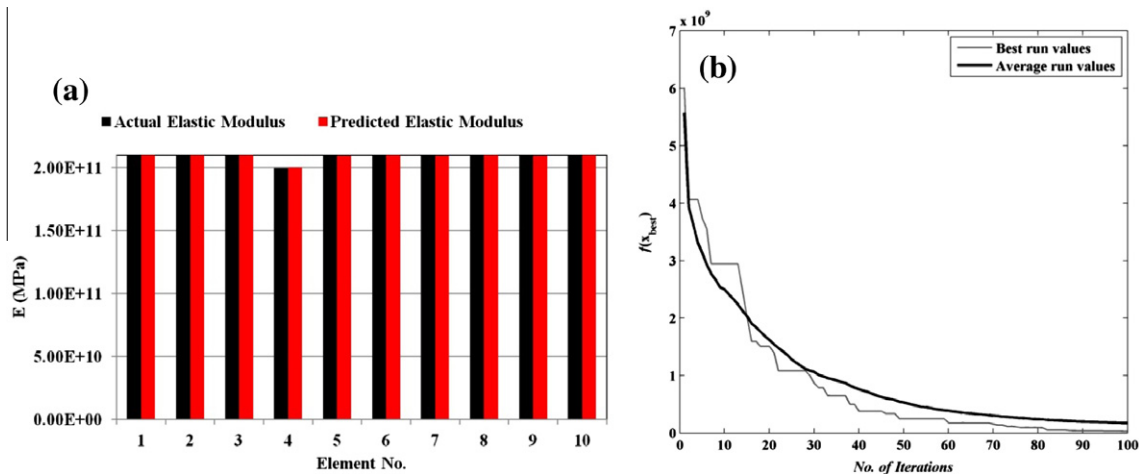


Fig. 4. Results of the 1st scenario. (a) Damage detection. (b) Convergence history.

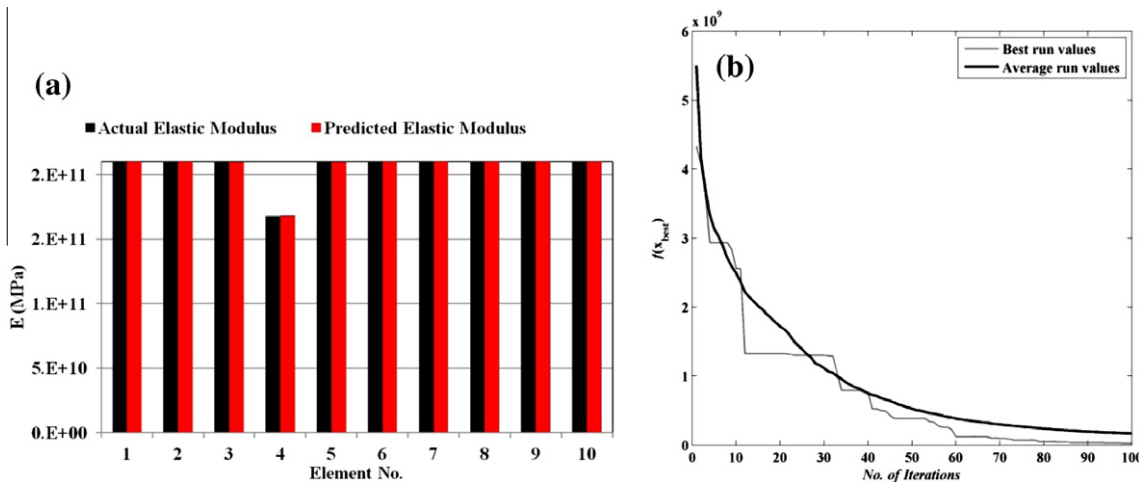


Fig. 5. Results of the 2nd scenario. (a) Damage detection. (b) Convergence history.

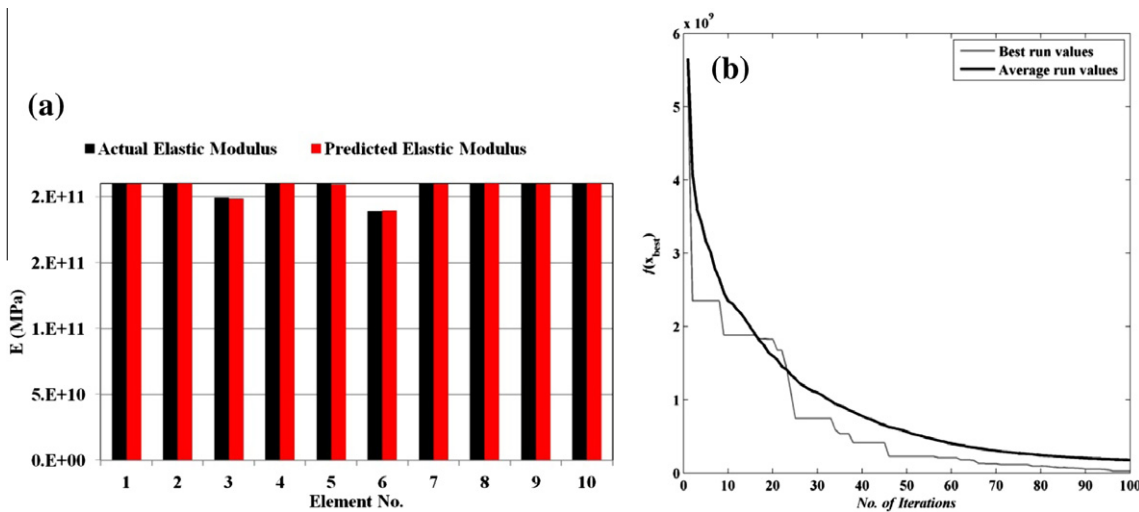


Fig. 6. Results of the 3rd scenario. (a) Damage detection. (b) Convergence history.

tively. As it is shown in these figures, the results confirm that the CAPSO is able to find the global optimums and locates the damage with a high degree of accuracy.

6.2. Gear train design

Gear train design problem is a discrete optimization problem. It was introduced by Sandgran [31] and it has four integer variables between 12 and 60. This problem consists of cost minimization of the gear ratio of a compound gear train (see Fig. 7). The gear ratio can be defined as:

$$\text{Gear ratio} = \frac{\text{Angular velocity of the output shaft}}{\text{Angular velocity of the input shaft}} \quad (33)$$

where T_i denotes the number of teeth of the i th gear wheel which should be integers. It is required to obtain the teeth numbers on the wheels so that a gear ratio reaches to $1/6.931$. Therefore, the mathematical formulation of the objective function is as follows:

$$f(T_a, T_b, T_d, T_f) = \left(\frac{1}{6.931} - \frac{T_b T_d}{T_a T_f} \right) \quad (34)$$

Table 8 shows the best results for the best objective value obtained in this study and also those of in the literature [31–40]. The obtained solution after 2000 searches by the proposed CAPSO is $(T_a, T_b, T_d, T_f) = (49, 19, 16, 43)$ with the objective value 2.7009×10^{-12} . As it can be seen from the table, the best result obtained in this study is better than the results obtained in [31–38] and same as the results obtained by Parsopoulos and Vrahatis [39] and Gandomi et al. [40]. Although, Parsopoulos

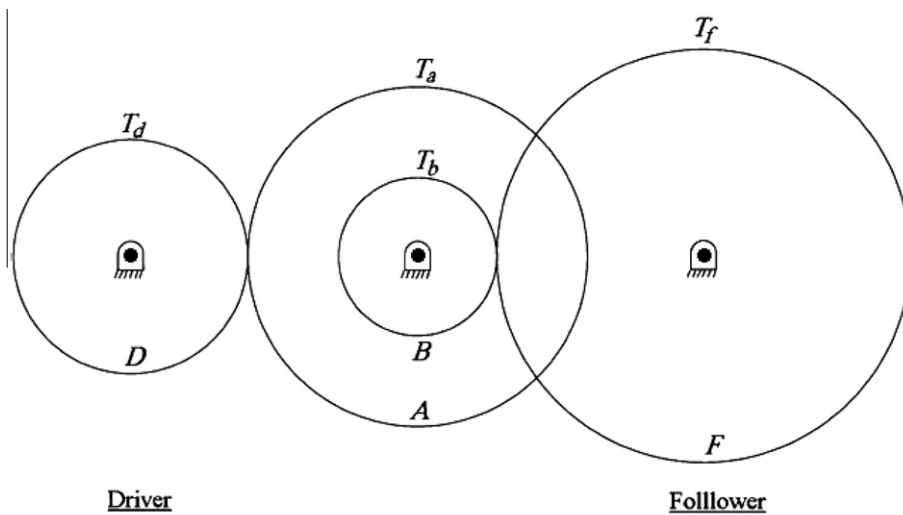


Fig. 7. Schematic representation of gear train.

Table 8
Optimal design comparison for the gear train.

Refs.	T_a	T_b	T_d	T_f	Gear ratio	$f_{\min}$
[31]	45	22	18	60	0.1467	5.712×10^{-6}
[32]	33	15	13	41	0.1441	2.146×10^{-8}
[33]	N/A ^a	N/A	N/A	N/A	N/A	2.33×10^{-7}
[34]	47	29	14	59	0.1464	4.5×10^{-6}
[35]	52	15	30	60	0.1442	2.36×10^{-9}
[36]	52	15	30	60	0.1442	2.36×10^{-9}
[37]	33	14	17	50	0.1442	1.362×10^{-9}
[38]	42	16	19	50	0.1447	0.23×10^{-6}
[39]	43	16	19	49	0.1442	2.701×10^{-12}
[40]	43	16	19	49	0.1442	2.701×10^{-12}
CAPSO	49	19	16	43	0.1442	2.701×10^{-12}

^a N/A is not available.

and Vrahatis [39] used PSO algorithm and obtained the same results, the maximum number of searches they used (100,000) is far more than the number of searches used here. The number of evaluation used hers is also less than the Gandomi's study [40] which is equal to 5000 (Table 8).

6.3. Pressure vessel design

The pressure vessel design is one of the benchmark structural engineering problems [41]. A schematic cylindrical pressure vessel capped at both ends by hemispherical heads is shown in Fig. 8. The air tank has a minimum volume of 750ft³ and a working pressure of 3000psi. Here, it is designed based on the American Society of Mechanical Engineers (ASME) boiler and pressure vessel code. The cost of single 60° welding and material and forming cost are to be minimized. There are four involved variables, including the thickness (T_s), thickness of the head (T_h), the inner radius (R), and the length of the cylindrical section of the vessel (L). The first two variables (thicknesses) are discrete values which are integer multiples of 0.0625 in.

The optimization problem of the pressure vessel design can be expressed as follows:

$$f(T_s, T_h, R, L) = 0.6224T_sRL + 1.7781T_hR^2 + 3.1661T_s^2L + 19.84T_h^2L \quad (35)$$

Subject to:

$$g_1 = -T_s + 0.0193R \leq 0 \quad (36)$$

$$g_2 = -T_h + 0.00954R \leq 0 \quad (37)$$

$$g_3 = -\pi R^2L - \frac{4}{3}\pi R^3 + 1296,000 \leq 0 \quad (38)$$

$$g_4 = L - 240 \leq 0 \quad (39)$$

where $1 \times 0.0625 \leq T_s, T_h \leq 99 \times 0.0625$ and $10 \leq R \leq 200$.

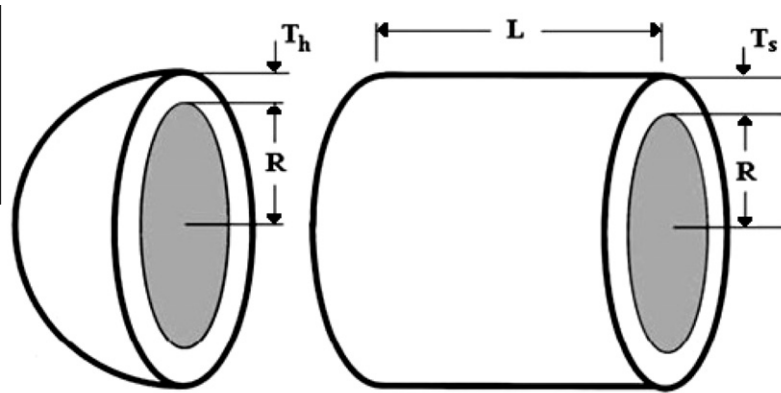


Fig. 8. Schematic representation of pressure vessel.

Table 9

Optimal design comparison for the pressure vessel.

Ref. Method	[42] HS	[43] SA	[44] FA	[45] GA	[46] GA-PSO	[47] PSO	CAPSO
Ts	0.75	0.7683	0.75	0.75	0.75	0.75	0.75
Th	0.375	0.3797	0.375	0.375	0.375	0.375	0.375
R	38.8601	39.80962	38.8601	38.858	38.8601	36.684	38.8601
L	221.3655	207.2256	221.3655	221.402	221.3655	224.096	221.3655
fmin	5849.762	5868.765	5850.383	5850.77	5850.383	5875.254	5850.383
g1	0	0	0	-4.06E-5	0	-0.042	-3.22E-9
g2	-0.0043	0	-0.0043	-4.29E-3	-0.0043	-0.02503	-0.00427
g3	0.2713^a	-10.7065	-0.0134	-19.663	0.0446	141806.5	-0.06867
g4	-18.6345	-32.7744	-18.6345	-18.598	-18.6345	-15.904	-18.6345

^a Bold sets are violated sets.

This problem has been solved by several researchers as a benchmark structural optimization problem. The comparisons of results for several approaches for this mixed variable problem using different algorithms are presented in Table 9. It is notable that the CAPSO requires 40 particles and 250 iterations to reach the optimum.

From Table 9, the results obtained by the CAPSO are better than PSO and other available solutions. Note that the best objective value obtained by HS is better than CAPSO but it is not feasible since the third constraint is violated. Although, the results of FA are very close to the CAPSO results, the FA evaluated the objective function 50,000 times which is five times more than the number of evaluation used in this study.

7. Discussions and conclusions

Chaos has been introduced to the standard APSO to develop the new chaotic APSO. Twelve different chaotic maps have been investigated to tune the main parameter, β , of the APSO. By comparing different chaotic APSO variants, the algorithm that uses the sinusoidal map as its β is the best chaotic APSO (CAPSO). The results have also revealed the significant improvement of the new algorithm with the application of deterministic chaotic signals in place of constant value. Statistical results and success rates of the CAPSO suggest that the tuned APSO clearly improves the reliability of the global optimality and they also enhance the quality of the results. In comparison with other chaotic PSO variants proposed in the literature, it seems the CAPSO performed significantly well. The results of CAPSO on three engineering problems show its applicability for the complex real-world problems.

In addition, it can be observed that the main advantage of the CAPSO is that there is a fewer number of parameters, compared with other PSO variants. All algorithms have algorithm-dependent parameters, and the performance of an algorithm is largely determined by these parameters. Therefore, it is often one of the most difficult tasks to find best values for parameters in an optimization algorithm. In this paper, we provided a solution to this challenging issue by using chaotic maps to replace these parameters. In CAPSO, there is only one parameter, β , to be tuned using chaos theory. The proposed approach has been then evaluated and tested using a complex engineering problem and the results clearly show that the CAPSO can solve the real-world problems efficiently.

Furthermore studies on convergence analysis can be fruitful. It can be expected that for different types of problem, different chaotic maps may have different convergence rates. In addition, further topics of studies can also focus on the extension of the CAPSO to solve mixed-type problems and discrete optimization problems.

Acknowledgement

Contributions by the first two authors to this research work are supported by the New Faculty Startup Fund from the University of Akron. The authors are grateful for this support.

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